Hamming Codes

- Hamming code uses redundant bits (extra bits)
- Hamming codes can be classified as
- Systematic --- Message bits and Parity bits can be identified as in the case of linear block codes
- Non systematic--- No identification between Message bits and Parity bits

Hamming Codes

- Hamming code uses redundant bits (extra bits) which are calculated according to the below formula:-
- **Step 1: To calculate the number of parity bits if data bits is only given**

$$2^r \geq m+r+1$$

- Where **r** is the number of redundant bits required and **m** is the number of data bits.
- **Step 2: To calculate the position of parity bits , Its calculated by 2^n where n=0,1,2,3**
- **R1** bit is appended at position $\mathbf{2^0}$
- **R2** bit is appended at position $\mathbf{2^1}$
- **R3** bit is appended at position $\mathbf{2^2}$ and so on.

- **Step 3: To calculate the value of parity bits , its done by the combination of Bit 1 at bit positions 1, 2, 3,…**

Hamming Codes- To detect error at the receiver

- These redundant bits are then added to the original data for the calculation of error at receiver’s end.
- At receiver’s end with the help of even parity (generally) the erroneous bit position is identified and since data is in binary we take complement of the erroneous bit position to correct received data.
- Respective index parity is calculated for **r1, r2, r3, r4** and so on.

Hamming Codes

- Hamming code uses redundant bits (extra bits) which are calculated according to the below formula:-
- **$2^r \geq m+r+1$**
- Where **r** is the number of redundant bits required and **m** is the number of data bits.
- **R** is calculated by putting **r = 1, 2, 3 …** until the above equation becomes true.
- **R1** bit is appended at position **2^0**
- **R2** bit is appended at position **2^1**
- **R3** bit is appended at position **2^2** and so on.

Hamming Codes

- Given the data 1001101, generate the hamming code
- **Check for the condition $2^r \geq m+r+1$**

r-= 1 does not satisfy the relation

r-= 2 does not satisfy the relation

r= 3 does not satisfy the relation

r= 4 does satisfy the relation

- **$2^r \geq 7+r+1$**
- **r = 4 ; 16 ≥ 7+4+1**
- **16 ≥12**

Hamming Codes

- **R** is calculated by putting **r = 1, 2, 3 …** until the above equation becomes true.
- **R1** bit is appended at position $\mathbf{2^0}$
- **R2** bit is appended at position $\mathbf{2^1}$
- **R3** bit is appended at position $\mathbf{2^2}$ and so on

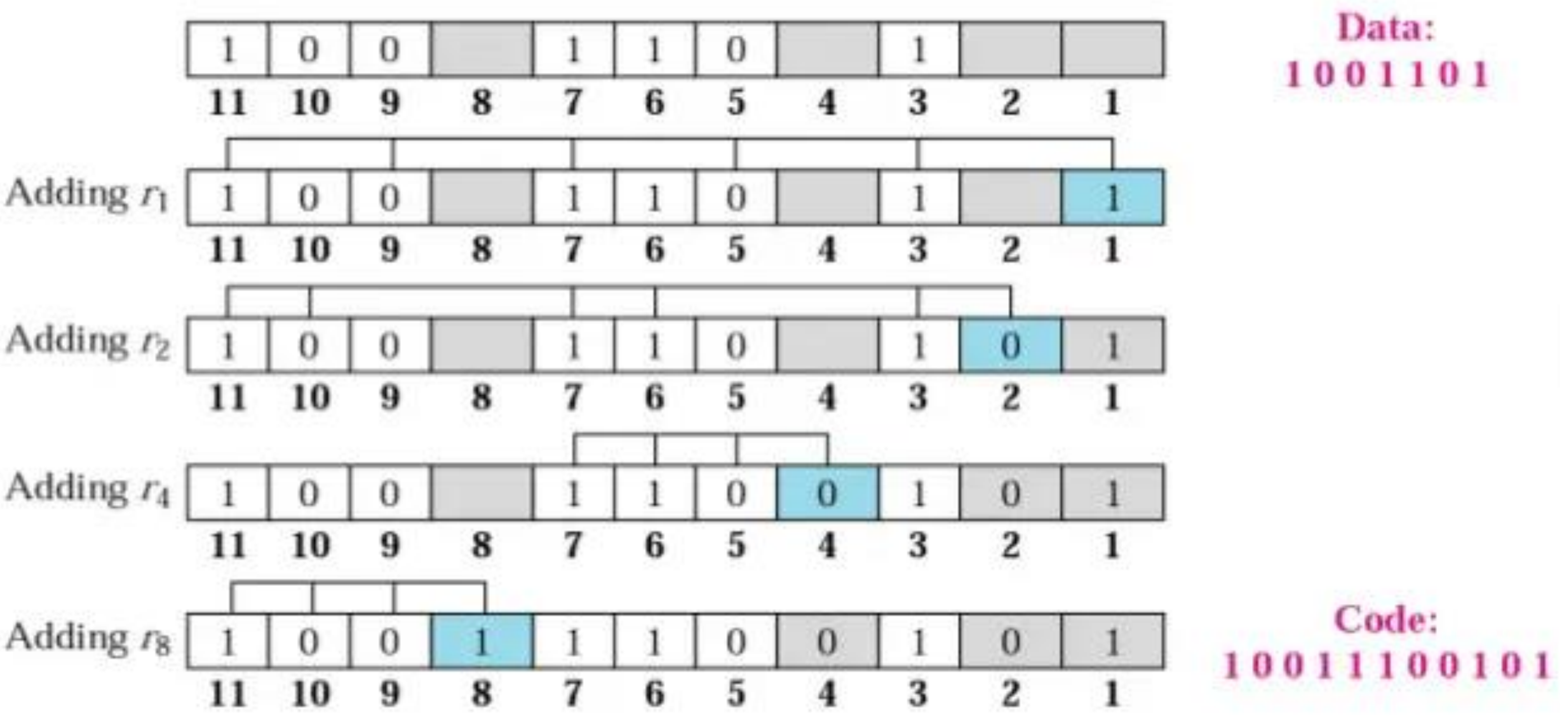
- **R1** bit is appended at position $\mathbf{2^0 = 1}$
- **R2** bit is appended at position $\mathbf{2^1 = 2}$
- **R3** bit is appended at position $\mathbf{2^2}$ $= 4$
- **R4** bit is appended at position $\mathbf{2^3}$ $= 8$

Hamming Codes

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

- R1(1,3,5,9,7,11)
- R2(2,3,6,7,10,11)
- R3(4,5,6,7)
- R4(8,9,10,11)

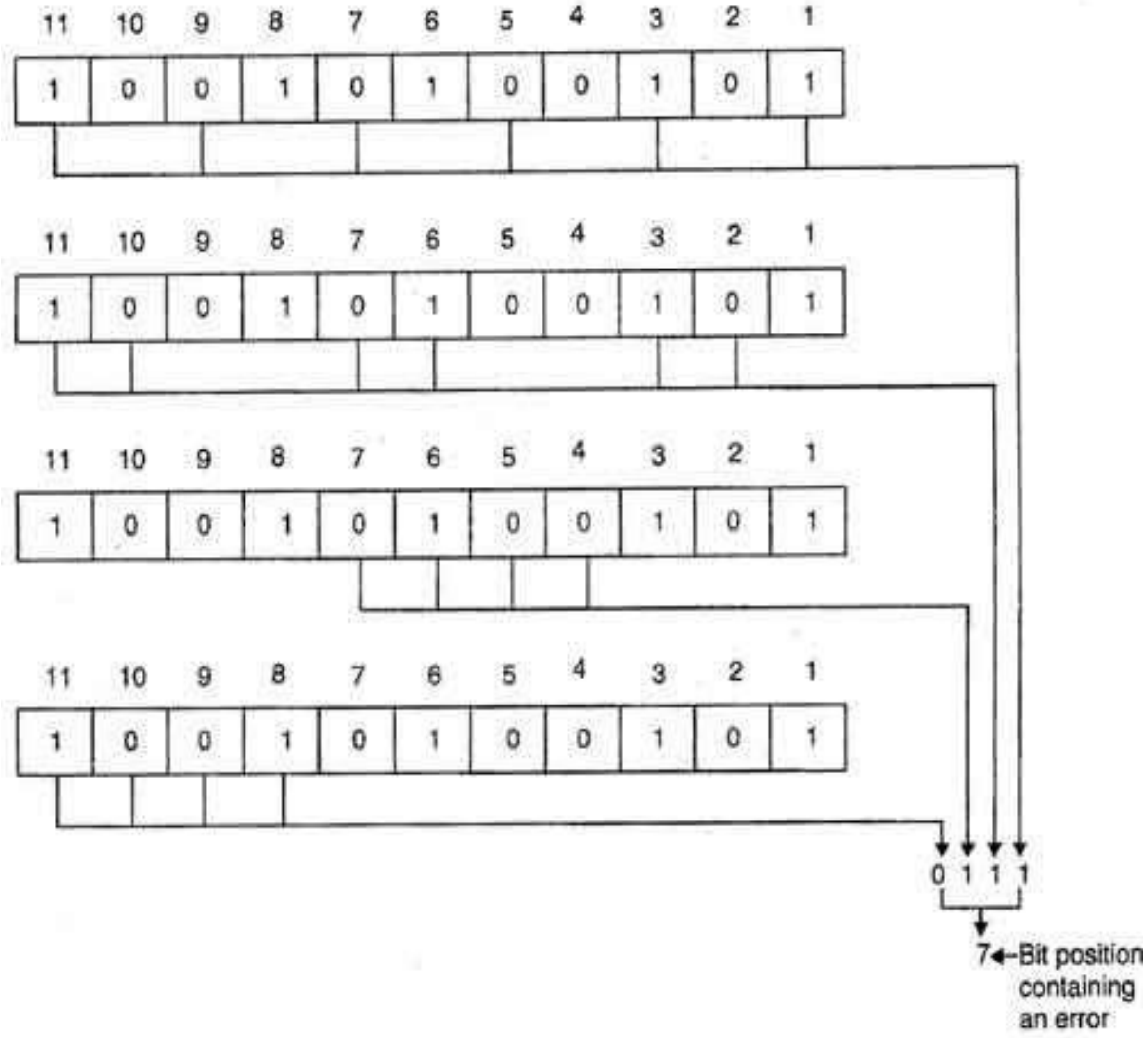
Example: *Hamming Code*



Thus data 1 0 0 1 1 1 0 0 1 0 1 with be transmitted.

Error Detection & Correction

Considering a case of above discussed example, if bit number 7 has been changed from 1 to 0.The data will be erroneous



Data sent: 1 0 0 1 1 1 0 0 1 0 1

Data received: 1 00 1 0 1 00 1 0 1 (seventh bit changed)

The receive takes the transmission and recalculates four new VRCs using the same set of bits used by sender plus the relevant parity (r) bit for each set as shown in fig.

Then it assembles the new parity values into a binary number in order of r position (r_8, r_4, r_2, r_1).

In this example, this step gives us the binary number 0111. This corresponds to decimal 7. Therefore bit number 7 contains an error. To correct this error, bit 7 is reversed from 0 to 1.

Example problem 2

Let us assume the even parity hamming code from the above example (111001101) is transmitted and the received code is (110001101). Now from the received code, let us detect and correct the error.

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Let us assume the even parity hamming code from the above example (111001101) is transmitted and the received code is (110001101). Now from the received code, let us detect and correct the error.

To detect the error, let us construct the bit location table.

Bit Location	9	8	7	6	5	4	3	2	1
Bit designation	D5	**P4**	D4	D3	D2	**P3**	D1	**P2**	**P1**
Binary representation	1001	1000	0111	0110	0101	0100	0011	0010	0001
Received code	1	1	0	0	0	1	1	0	1

Checking the parity bits

For P1 : Check the locations 1, 3, 5, 7, 9. There is three 1s in this group, which is wrong for even parity. Hence the bit value for P1 is 1.

For P2 : Check the locations 2, 3, 6, 7. There is one 1 in this group, which is wrong for even parity. Hence the bit value for P2 is 1.

For P3 : Check the locations 3, 5, 6, 7. There is one 1 in this group, which is wrong for even parity. Hence the bit value for P3 is 1.

For P4 : Check the locations 8, 9. There are two 1s in this group, which is correct for even parity. Hence the bit value for P4 is 0.

The resultant binary word is 0111. It corresponds to the bit location 7 in the above table. The error is detected in the data bit D4. The error is 0 and it should be changed to 1. **Thus the corrected code is 11100 110 1.**

Practice Problems

Consider a 7-bits even parity Hamming code received as 1100010. Check for the correctness of data received

Solution

In 7-bits Hamming code **4-bits (D3, D5, D6 & D7)** represents ***data bits*** and **3-bits (P1, P2, & P4)** are ***parity bits.***

This 7-bit Hamming code is represented as follows:

Bit Position	1	2	3	4	5	6	7
Hamming Code Representation	**P1**	**P2**	**D3**	**P4**	**D5**	**D6**	**D7**
Received Hamming Code	1	1	0	0	0	1	0

Practice Problems

Bit Position	1	2	3	4	5	6	7
Hamming Code Representation	**P1**	**P2**	**D3**	**P4**	**D5**	**D6**	**D7**
Received Hamming Code	1	1	0	0	0	1	0

Practice Problems

- To check whether the received Hamming code is correct or not let's perform the following operations:
- Take the bits present at the below-mentioned places
- Parity bit-1 covers all the bits positioned at bit-position 1, 3, 5, 7, 9, 11,.. etc.
- Therefore,
- C1 = Even_Parity(1, 3, 5, 7) = 1 0 0 0 = 1 (Odd number of 1,s, which means error is present)
- Parity bit-2 covers all the bits positioned at 2, 3, 6, 7, 10, 11,... etc.
- Therefore,
- C2 = Even_Parity(2, 3, 6, 7) = 1 0 1 0 = 0 (Even number of 1's, which means No error is present)
- Parity bit-4 covers all the bits positioned at 4–7, 12–15, 20–23,... etc.
- Therefore,
- C4 = Even_Parity(4, 5, 6, 7) = 0 0 1 0 = 1 (Odd number of 1's, which means error is present)
- Therefore,
- C = C4 C2 C1 = 1 0 1 = It represent decimal number 5

Practice Problems

- That means a Single bit error is present at the 5th
- position of received Hamming code 1100010.
- Now, remove the 5th
- position single bit error.

- The 5th bit holds 0 which creates an error in a Hamming encoded 7-bit number. Hence, to remove this error 0 is replaced with 1.

Hamming code - ECC

→ one of the most
and correct error.

→ Most commonly used

→ Easy to implement

Types

→ 7 bit hamming code < 4 bits data, 3 bit parity

→ 12 bit hamming code < 8 bits data, 4 bits parity

→ 15 bit hamming code < 11 bits data, 4 bits parity

7 bit hamming Code

$\rightarrow$ 2^n ($n = 0, 1, 2, 3 \cdots$)

n = number of parity bits

$n = 3$

$2^0, 2^1, 2^2$

$\downarrow \quad \downarrow \quad \downarrow$

$1 \quad 2 \quad 4$

→ Bit

Bit designation	D_7	D_6	D_5	P_4	D_3	P_2	P_1
Bit location	7	6	5	4	3	2	1
Binary location number	111	110	101	100	011	010	001

Bit designation	D_7	D_6	D_5	P_4	D_3	P_2	P_1
Bit location	7	6	5	4	3	2	1
Binary location number	111	110	101	100	011	010	001

P_1 : 1, 3, 5, 7

P_2 : 2, 3, 6, 7

P_4 : 4, 5, 6, 7

Assume that the even parity

0110011 is transmitted and that 0100011 is

eceived. Determine bit location where error

as occured using received data.

u:

it designation	D_7	D_6	D_5	P_4	D_3	P_2	P_1
t location	7	6	5	4	3	2	1
ary location	111	110	101	100	011	010	001

Bit designation	D_7	D_6	D_5	P_4	D_3	P_2	P_1
Bit location	7	6	5	4	3	2	1
Binary location number	111	110	101	100	011	010	001
Received data	0	1	0	0	0	1	1

$$\left.\begin{matrix} P_1 : 1, 3, 5, 7 \\ : 1, 0, 0, 0 \end{matrix}\right\} \quad P_1 \rightarrow 1$$

$$\left.\begin{matrix} P_2 : 2, 3, 6, 7 \\ 1, 0, 1, 0 \end{matrix}\right\} \quad P_2 \rightarrow 0$$

$$4, 5, 6, 7 \;\} \quad P_4 \rightarrow 1$$

$$(101)_2 = (5)_{10}$$

Soln :

Bit designation	D_7	D_6	D_5	P_4	D_3	P_2	P_1
Bit location	7	6	5	4	3	2	1
Binary location number	111	110	101	100	011	010	001
Received data	0	1	0	0	0	1	1

P_1 : 1, 3, 5, 7
: 1, 0, 0, 0 } $P_1 \rightarrow 1$

P_2 : 2, 3, 6, 7
1, 0, 1, 0 } $P_2 \rightarrow 0$

P_4 : 4, 5, 6, 7
0, 0, 1, 0 } $P_4 \rightarrow 1$

$(101)_2 = (5)_{10}$

Correct code is 0110011

Problem: A 7 bit Hamming Code is received as 1011011. Assume Even parity & state whether the received Code is correct or wrong. if wrong locate the bit in error.

D_7 D_6 D_5 P_4 D_3 P_2 P_1

1 0 1 1 0 1 1

...ting errors!

Step 1: Analyzing bits 1, 3, 5, ~~2~~ ~~6~~ 7

we have $P_1\ D_3\ D_5\ D_7 \implies 1011$ $\left\{ \begin{array}{l} \text{odd parity} \\ \text{error exits} \end{array} \right.$

We put $P_1 = 1$

$\Rightarrow 1\ 0\ 0\ 1$ $\left\{ \begin{array}{l} \text{even parity} \\ \text{no errors} \end{array} \right\}$

$P_2 = 0$

Step 3: analyzing bits 4, 5, 6 & 7

we have $P_4\ D_5\ D_6\ D_7$

$\Rightarrow 1\ 1\ 0\ 1$ $\left\{ \begin{array}{l} \text{odd parity} \\ \text{error exists} \end{array} \right\}$

we put $P_4 = 1$

error word E =

P_4	P_2	P_1
1	0	1

decimal value $E = 5$ which shows that the 5^{th} bit is in error.

so, we write the correct word by simply inverting the 5^{th} bit.

$\therefore$ Correct word = 1001011

Practice Problems

1. Calculate the number of redundancy bits r required to correct n bits of data. Apply your result for the case n = 15.

2. Use the Hamming code to calculate the codeword for 10011010. Explain each stage in your calculation.

3. Use the Hamming code to calculate the codeword for 10011010. Assume that instead of 111100101010 the receiver got 111100101110. Show how the receiver can calculate which bit was wrong and correct it.

4. Test if code word 010101100011 is correct, assuming it was created using an even parity Hamming code. If it is incorrect, indicate what the correct code word should have been. Finally, indicate what the original data was transmitted

Block Codes

- Data Word is 0100 Code Word is 0100011
- Syndrome = 000 What is the code word at receiver?

- Data Word is 0111 code Word is 0111001
- Syndrome = 011 What is the code word at receiver?

- Data Word is 1101 Code Word is 1101000
- Syndrome : 101 What is the code word at receiver?

Points in a nutshell

Error control method		
Check sum	Check sum = 0	Retransmission , no error detected
LRC	Even parity bits	If Even parity bits is not received single bit errors can be detected
VRC	Even parity bits	If Even parity bits is not received single bit errors can be detected
Block codes	Systematic- Hamming codes	Syndrome $S= YH^T$; S= 0 implies no error in the received vector S= non zero implies error in the received bit S value tells particular bit is in error Eg:101 tells that 5 th bit is in error
	Non Systematic –Hamming codes	As per the procedure
CRC	CRC Division = zero	0 implies no error in the received vector non zero implies error in the received bit

- Linearity Property
- Cyclic property
- Hamming distance
- Hamming Weight
- Properties of linear block codes
- Properties of Cyclic codes
- Properties of G(p)
- Disadvantages of LRC &VRC

Why cant apply Hamming distance?

How will u decide r bits?

Why XOR instead of AND (or) OR operation?

How will u know the location of error in Hamming code algorithm?

How rearranging the data makes u to detect burst error